Course Title	Mobile Application Development		
Course ID	CS2703	**Course Type**	CS Elective
Credit hours	3 (2+1)	**Hours per week (C-L)**	3-3
Program(s)	ADP Computing CS	**Preferred Semester**	4th

Weekly Lab Plan:

Week	Contents	Activities
Week 1	**Introduction to Mobile Application Development** ♦ Course overview and objectives ♦ Importance of mobile app development ♦ Overview of mobile platforms (Android, iOS, Cross-Platform) ♦ Setting up the development environment ♦ Installing IDEs (Android Studio, Xcode, VS Code) ♦ Introduction to Flutter and React Native	**Lab Task:** 1. Install **Flutter SDK** and set up Android Studio/VSCode. 2. Create a **"Hello World" app** in Flutter and run it on an emulator/device. 3. Explore **React Native** setup using expo-cli.
Week 2	**Fundamentals of Mobile App Development** ♦ Introduction to Flutter ♦ Widgets, layouts, and state management ♦ Introduction to React Native ♦ Components, props, and state	**Lab Task:** 1. Build a **Flutter app** with: ○ A Stateless Widget displaying a profile card. ○ A StatefulWidget with a counter button. 2. Create a **React Native app** with: ○ A functional component displaying a list. ○ A stateful counter component
Week 3	**User Interface (UI) Design** ♦ Principles of UI/UX design ♦ Material Design (Android) and Human Interface Guidelines (iOS) ♦ Building responsive UIs ♦ Using Flutter widgets and React Native components	**Lab Task:** Design a login screen in Flutter using MaterialApp widgets (TextField, ElevatedButton). Recreate the same login screen in React Native with StyleSheet. Compare Material Design

		(Android) and Cupertino (iOS) widgets.
Week 4	**Navigation and Routing** ♦ Navigation in Flutter ♦ Navigator, routes, and named routes ♦ Navigation in React Native ♦ React Navigation library	**Lab Task:** Implement multi-screen navigation in Flutter using Navigator. Set up tab navigation in React Native with react-navigation. Pass data between screens (e.g., user profile → details page).
Week 5	**State Management** ♦ State management in Flutter ♦ setState, Provider, Riverpod ♦ State management in React Native ♦ useState, useContext, Redux	**Lab Task:** Use Provider in Flutter to manage a shopping cart state. Implement Redux in React Native for a todo list app. Compare state management approaches.
Week 6	**Working with APIs** ♦ RESTful APIs and JSON ♦ Making HTTP requests (GET, POST, PUT, DELETE) ♦ Integrating APIs in Flutter and React Native ♦ Using packages like http (Flutter) and axios (React Native)	**Lab Task:** Fetch data from a REST API (e.g., JSONPlaceholder) in Flutter using http. Display the data in a ListView.builder. Repeat in React Native using axios.
Week 7	**Database Integration** ♦ Local storage options ♦ SQLite, SharedPreferences (Flutter), AsyncStorage (React Native) ♦ **Firebase integration** ♦ **Firebase Firestore, Realtime Database, and Authentication**	**Lab Task:** Use Firebase Firestore to: Save user data from a form. Retrieve and display data in a Flutter/React Native app.

		Implement authentication with Firebase Auth.
Week 8	**Advanced UI Components** ♦ Custom widgets and animations in Flutter	**Lab Task:** Create a custom animated widget in Flutter (e.g., a loading spinner). Build a swipeable card in React Native using react-native-gesture-handler.
	♦ Custom components and animations in React Native	
Week 9	**Mid Term Examination**	
Week 10	**App Testing and Debugging** ♦ Unit testing and widget testing in Flutter ♦ Testing in React Native ♦ Jest and React Native Testing Library	**Lab Task:** Write unit tests for a Flutter counter app. Debug a React Native app using React DevTools. Test apps on physical devices.
Week 11	**App Deployment** ♦ Preparing apps for deployment ♦ Building APKs (Android) and IPAs (iOS) ♦ Publishing apps on Google Play and Apple App Store	**Lab Task:** Generate a release APK for a Flutter app. Build an IPA for iOS (simulator only, unless Mac available). Publish a demo app to GitHub Pages (for web).
Week 12	**Cross-Platform Development** ♦ Advantages and challenges of cross-platform development ♦ Comparing Flutter and React Native	**Lab Task:** Benchmark performance of the same app in Flutter vs. React Native. Discuss pros/cons of each framework.
Week 13	**Performance Optimization** ♦ Optimizing app performance in Flutter	**Lab Task:** Encrypt sensitive data using flutter_secure_storage. Secure API calls with JWT tokens in React Native.

	♦ Optimizing app performance in React Native	
Week 14	**Security in Mobile Apps** ♦ Secure coding practices ♦ Data encryption, secure storage, and secure API calls ♦ Handling permissions and user privacy	**Lab Task:** Convert a Flutter app to a PWA (Progressive Web App). Integrate a pre-trained ML model (e.g., TensorFlow Lite) for image classification.
Week 15	**Emerging Trends in Mobile Development** ♦ Introduction to Progressive Web Apps (PWAs) ♦ Exploring AI and ML in mobile apps	**Lab Task:** Develop a complete mobile app (e.g., e-commerce, weather, or social app). **Include:** User authentication API/database integration Responsive UI
Week 16	**Final Project Development** ♦ Project planning and design ♦ Defining app features and architecture ♦ Implementing core features of the final project	
Week 17	♦ Revision & Final Presentations ♦ Revision & Final Presentations	
Week 18	**FINAL TERM EXAM**	